The Temporal Blind Spot in Video Retrieval: Diagnosing Temporal Sensitivity

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Abstract

Temporal blindness in video models is well-documented for generative QA, but its consequences for *retrieval embeddings*, where failures are silent, remain uncharacterized. We diagnose temporal sensitivity across six retrieval benchmarks spanning copy detection, temporal manipulation, and high-overlap motion retrieval. A *temporal scramble gradient* cleanly separates order-invariant methods from sequence-aware ones. A feature-vs-comparator decomposition shows that on Honda HDD, 89% of the performance gap comes from the comparator (cosine→DTW), not the feature. On nuScenes, the comparator alone surpasses the residual transform (AP = 0.867 vs. 0.815). Embedding probes across three Video-LLMs and Claude 4.6 Opus localize where temporal signal is lost: per-frame vision features achieve perfect forward/reverse separation under sequence-aware comparators (by construction), but mean-pooling erases this signal ($s_{\text{rev}} \approx 1.0$), and neither linear nor nonlinear probes on LLM hidden states recover temporal order above chance (126 configurations tested). No method spans both copy detection and motion retrieval.

1 Introduction

Modern video retrieval systems represent videos as mean-pooled frame-level embeddings from pretrained vision encoders. This *bag-of-frames* (BoF) approach excels at standard Video Copy Detection (VCD), where the goal is to identify re-encoded, cropped, or camcorded versions of a video, because visual semantics are preserved regardless of frame order, enabling billion-scale nearest-neighbor search [20, 19].

However, BoF representations discard temporal order entirely. A growing literature documents this "temporal blindness" in Video-LLMs [45, 46, 42, 52, 53, 50]: generative probes show near-chance temporal QA and frame shuffling leaves benchmark scores intact. Yet this diagnosis has focused on generative question-answering, and the consequences for *retrieval embeddings*, where the downstream cost is silent failure, remain uncharacterized. We fill this gap with three diagnostic test cases:

1. Copy Detection & Scene Retrieval: When the discriminative signal is the visual background (e.g., standard VCD, or retrieving activities in different rooms), semantic pooling is highly effective.

2. Temporal Manipulation (Trust & Safety): Reversed video yields BoF similarity of 1.0, evading embedding-based copy retrieval entirely [7].

3. High-Overlap Motion Retrieval (Dashcam Fleets): Querying an autonomous fleet for "left turns at Intersection X" requires encoding motion dynamics, since the static background dominates the embedding, making left and right turns indistinguishable under BoF.

We evaluate these cases using DINOv3 [2] (BoF and DTW attention trajectories), V-JEPA 2 [1] (spatiotemporal prediction residuals), ViCLIP [58] (video-native contrastive baseline), and three Video-

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LLMs (Qwen3-VL [13], Gemma 4 [14], LLaVA-Video [15]), across six benchmarks: VCDB [3], Nymeria [32], MUVR [18], Honda HDD [4], nuScenes [5], and EPIC-Kitchens-100 [6].

Contributions. (1) A *temporal scramble gradient* that continuously stress-tests temporal sensitivity in retrieval embeddings, cleanly separating method classes on VCDB. (2) A *feature-vs-comparator decomposition* and embedding probes across the VLM pipeline that localize where order signal is lost (on HDD, 89% in the comparator, not the encoder), extending the temporal blindness diagnosis from generative QA to retrieval embeddings. (3) A controlled sampling-density analysis on Honda HDD that characterizes how the residual advantage scales with temporal dilution.

2 Methods

Observation 1 (Symmetric aggregation implies permutation-invariance)**.** *Any function $g(f_1, \ldots, f_T)$ satisfying $g(f_1, \ldots, f_T) = g(f_{\pi(1)}, \ldots, f_{\pi(T)})$ for all permutations π is order-invariant by construction. Mean-pooling and max-pooling are symmetric aggregators; Chamfer matching is a symmetric pairwise comparison. BoF order-invariance follows directly from the aggregation operator, independent of encoder capacity or training objective.*

DINOv3 & Attention Trajectories. We extract 1024-d CLS embeddings per frame from DINOv3 ViT-L [2]. *BoF* mean-pools frame embeddings into a single vector, compared via cosine similarity. *Chamfer similarity* [26] computes the pairwise frame similarity matrix and averages the symmetric maximum: $\frac{1}{2}\big(\mathrm{mean}_i \max_j S_{ij} + \mathrm{mean}_j \max_i S_{ij}\big)$, which is order-agnostic but retains per-frame structure. For structural tracking, we compute the spatial center-of-mass of the CLS→patch attention map (last layer, averaged over all heads) per frame, yielding a 2D trajectory compared via DTW. Temporal derivatives of CLS embeddings capture frame-to-frame transition dynamics, also compared via DTW.

V-JEPA 2 Temporal Residuals. V-JEPA 2 ViT-L ingests 64 frames as $32 \times 16 \times 16$ spatiotemporal patches ($D = 1024$). The *prediction residual* $R = \hat{P} - P$ (predicted minus actual masked patches) captures motion-specific error patterns. We spatially average each temporal position to obtain $\bar{R} \in \mathbb{R}^{T_{\text{mask}} \times D}$, compared via DTW with similarity $\exp(-d_{\text{DTW}})$. The *bag-of-tokens* (BoT) baseline mean-pools all encoder patches into a single vector (cosine similarity).

Temporal Scramble Gradient. We split video B's frame embeddings into K equal chunks, randomly shuffle them, and reassemble. Sweeping $K \in \{1, \ldots, 16\}$ traces a gradient from intact to fully scrambled. We shuffle *extracted* embeddings post-extraction (see Appendix M). Order-invariant methods should remain flat; sequence-aware methods should degrade.

VLM Generative Probes. We present forward and reversed clips to VLMs, classify temporal direction with three prompts, and report balanced accuracy.

ViCLIP. ViCLIP ViT-L [58], trained contrastively on 10M video clips, jointly encodes 8 frames via temporal attention into a single 768-d embedding (cosine similarity).

VLM Embedding Probes. Beyond generative classification, we extract intermediate representations from VLM pipelines: (1) *vision pooled*, the mean-pooled vision tower output (BoF over frame×patch tokens); (2) *vision sequence*, per-frame pooled vision tokens as a (T, D) sequence, compared via temporal derivative DTW; (3) *LLM hidden state*, the mean-pooled last-layer hidden state of the language backbone over all input tokens (vision + text). We measure s_{rev}, the similarity between forward and reversed representations of the same clip:

$$s_{\text{rev}} = \begin{cases} \cos(\mathbf{z}_{\text{fwd}},\ \mathbf{z}_{\text{rev}}) & \text{pooled probes (single vector),} \\ \exp\big(-\alpha \cdot d_{\text{DTW}}(\mathbf{Z}_{\text{fwd}},\ \mathbf{Z}_{\text{rev}})\big) & \text{sequence probes (temporal derivative DTW),} \end{cases} \tag{1}$$

where α follows the DTW α schedule defined below. $s_{\text{rev}} \approx 1.0$ indicates order-invariance; lower values indicate order sensitivity. Raw s_{rev} values exceeding 1.0 (from f32 cosine precision) are clamped to 1.000 in all tables and figures; affected entries are marked †. Because the two cases use different similarity functions, pooled and sequence s_{rev} values are not directly comparable (see Discussion).

DTW Implementation. All DTW comparisons use unconstrained warping with L2 cost matrices, path-length normalization ($d/(T_1 + T_2)$), and GPU-vectorized anti-diagonal wavefront process-

ing [22]. Similarity is $\exp(-\alpha \cdot d_{\text{DTW}})$ with $\alpha = 1$ for derivatives/residuals, $\alpha = 5$ for attention trajectories (min-max normalized to $[0, 1]$).

3 Experiments and Results

Evaluation protocols. All evaluations are pairwise (no query/gallery split). *VCDB*: 528 videos; positive pairs from annotations (5,585 unique), negatives sampled 1:1 with fixed seed; a sensitivity analysis across 1:1, 1:5, and all-pairs ratios shows stable rank order (Appendix H). AP/AUC via `sklearn.average_precision_score`. *HDD*: all $\binom{n}{2}$ pairs within each GPS-clustered intersection (DBSCAN, $\epsilon = 0.0003° \approx 30\,\text{m}$); positive = same maneuver label, negative = different; natural class ratio; 50 mixed clusters, 1,687 maneuvers. *nuScenes*: same protocol as HDD with $\epsilon = 30\,\text{m}$ on ego_pose; CAN bus–derived labels; 264 segments, 950 pairs. *EPIC* s_{rev}: per-sequence self-comparison ($s_{\text{rev}} = \text{sim}(\text{fwd}, \text{rev})$); 500 sequences; bootstrap 95% CIs with 1,000 resamples. *Nymeria*: 100 egocentric sessions, 8,780 segments; positive = same activity in same room; all $\binom{n}{2}$ pairs (394,210). *MUVR News*: 9,958 videos across 74 topics; positive = same topic (27,453 pairs).

3.1 VCDB Temporal Manipulation Diagnostics

On standard copy detection, BoF and Chamfer matching dominate: DINOv3 Chamfer achieves AP = 0.989 (528 videos, 5,585 positive pairs), while temporal methods underperform due to intra-class noise in frame-to-frame dynamics. The same ranking holds on Nymeria and MUVR (Appendix C).

This order-invariance is also a vulnerability: reversed video yields identical BoF similarity to the original. We measure this across VCDB positive pairs:

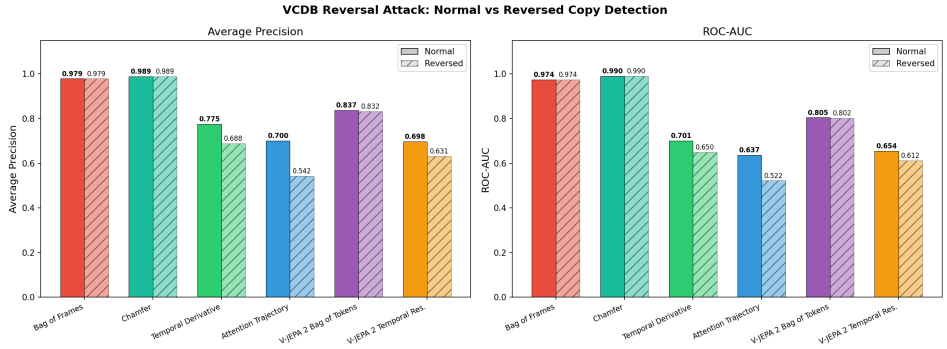

Figure 1: VCDB Reversal Attack (528 videos, 5,585 positive pairs). Order-invariant methods (BoF, Chamfer) maintain identical AP/AUC under reversal. Only sequence-aware methods (Attention Trajectory) register the structural tampering via an AP drop.

Table 1: VCDB Reversal Attack: 5,585 positive pairs (AP drop indicates detection of evasion).

Method	**Normal AP**	**Reversed AP**	**AP Δ**
Chamfer (DINOv3)	0.989	0.989	**-0.000** (Invariant)
BoF (DINOv3)	0.979	0.979	**-0.000** (Invariant)
BoT (V-JEPA 2)	0.838	0.832	-0.006
Attention Trajectory (DINOv3)	0.700	0.542	**-0.158** (Detects)
Temporal Residual (V-JEPA 2)	0.698	0.631	-0.067

BoF and Chamfer yield exactly zero AP delta under reversal. Only attention trajectory DTW clearly flags the reversed copy ($\Delta = -0.158$); temporal residual also registers a drop ($\Delta = -0.067$). To quantify temporal sensitivity beyond this binary test, we apply chunk-shuffle scrambling at increasing granularity ($K \in \{1, \ldots, 16\}$) across VCDB.

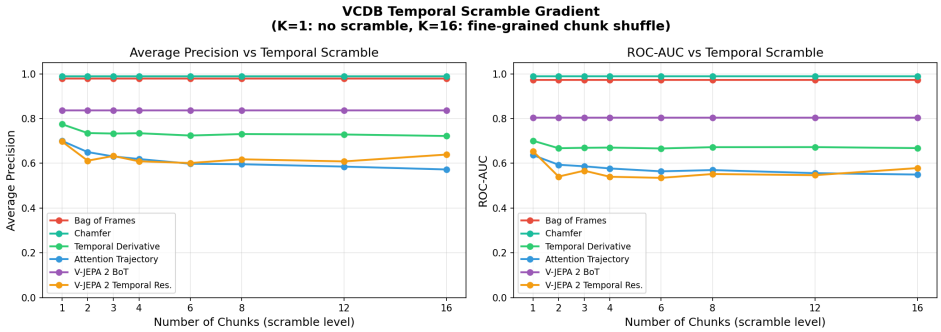

Figure 2: Temporal Scramble Gradient on VCDB (527 of 528 videos; one decode failure; 5,585 positive pairs). Order-invariant methods (BoF, Chamfer, V-JEPA 2 BoT) maintain constant AP regardless of scramble severity. Sequence-aware methods show overall degradation, with attention trajectories declining most steeply (-18.3% relative); V-JEPA 2 residual exhibits non-monotonic variation, possibly due to boundary effects at chunk boundaries (Appendix M).

The scramble gradient separates method classes (Figure 2; full table in Appendix M): order-invariant methods show no sensitivity, while attention trajectories degrade most (-18.3% relative from $K=1$ to $K=16$), as expected since gaze paths are the most temporally structured signal.

3.2 High-Overlap Motion: Honda HDD & nuScenes

To isolate motion from appearance, we clustered 128 Honda HDD [4] driving sessions by RTK GPS, extracting 1,687 maneuvers at 50 mixed-direction intersections where BoF similarity averages 0.63 ± 0.18 regardless of turn direction (Appendix J). The full dataset is passage-dominated ($\sim 66\%$), so AP is partially inflated by easier passing-vs-turn negatives.

Table 2: Honda HDD Maneuver Discrimination: 1,687 maneuvers across 128 sessions, 50 mixed-direction intersection clusters (same vs. different turn at same intersection).

Method	AP	AUC
DINOv3 Temporal Deriv.	0.498	0.526
DINOv3 Attn. Trajectory	0.507	0.541
DINOv3 BoF	0.530	0.542
DINOv3 Chamfer	0.559	0.577
V-JEPA 2 BoT	0.825	0.836
V-JEPA 2 Encoder-Seq DTW	0.942	0.935
V-JEPA 2 Temporal Residual	**0.956**	**0.952**

DINOv3 methods fail (AP $\approx$ 0.50): dashcam attention tracks salient objects identical for both turns. V-JEPA 2 temporal residuals achieve AP = 0.956; a pixel-difference motion proxy rules out a trivial intensity shortcut (Spearman $r = 0.051$; Appendix G), and optical flow (RAFT) achieves only AP = 0.478. Restricting to left-vs-right pairs only yields AP = 0.960 [0.958, 0.963]. To disentangle feature from comparator, we compare encoder patches via DTW: encoder-sequence DTW achieves AP = 0.942, showing that 89% of the BoT-to-residual gap (0.825 $\rightarrow$ 0.956) comes from the comparator (cosine $\rightarrow$ DTW). We restrict encoder-sequence DTW to motion tasks (HDD, nuScenes); on VCDB, BoF already achieves AP = 0.989.

3.2.1 Sampling Density: Context Window Sweep

The FPS downsample experiment (Appendix A) confounds sampling density with padding. We fix the backbone and pair set on HDD and sweep clip duration via `context_sec` $\in \{1.0, 2.0, 3.0, 4.0, 6.5, 8.0\}$ while always extracting 64 unique frames ($\text{fps}_{\text{eff}} = 64/d$).

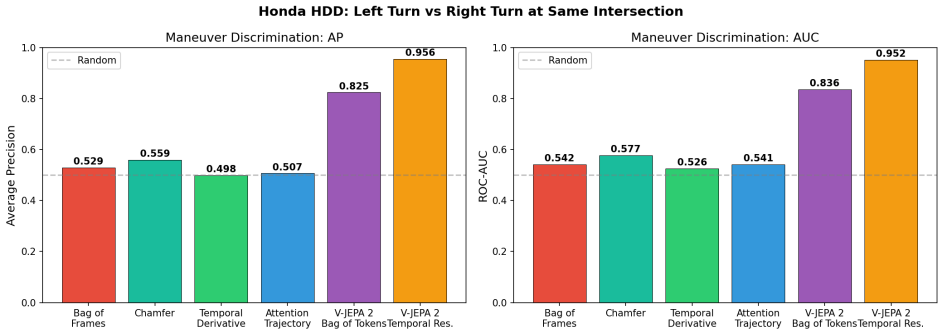

Figure 3: Honda HDD maneuver discrimination (1,687 maneuvers, 128 sessions). V-JEPA 2 temporal residuals achieve near-perfect separation (AP = 0.956) while DINOv3 methods remain at chance.

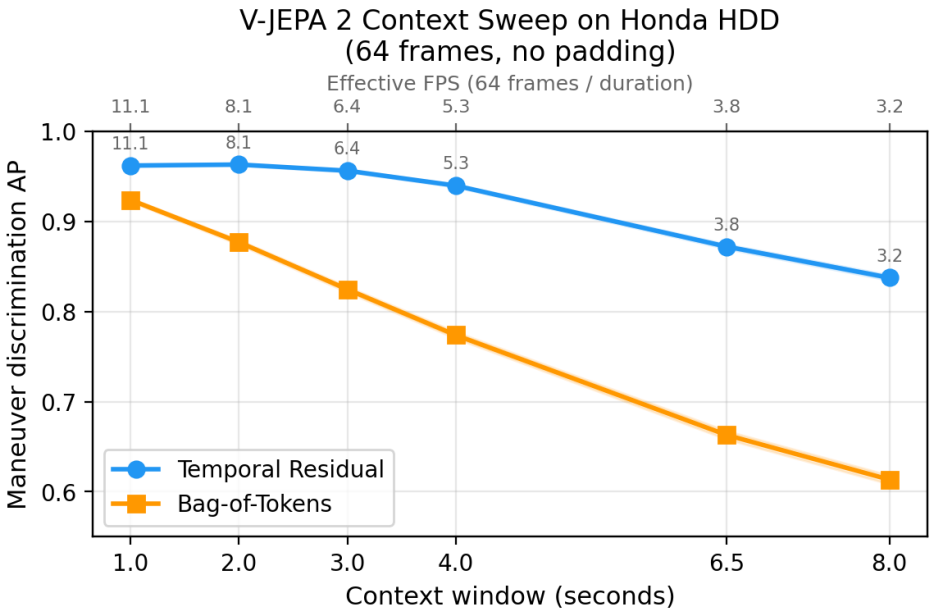

Figure 4: V-JEPA 2 context sweep on Honda HDD (1,687 maneuvers, 128 sessions). Temporal residual remains substantially more robust than BoT across all settings. The "knee" around 4–6.5 s marks a regime change where both signals degrade sharply.

Both signals degrade as the window widens, but temporal residual degrades more slowly. The gap grows with sparser sampling ($+0.224$ AP at 8 s): the residual holds up under temporal dilution where spatial content alone does not.

3.2.2 Cross-Dataset Validation: nuScenes

To test whether the method ranking transfers to a different platform, we evaluate on nuScenes [5], a different vehicle, camera, and environment. Turn labels come from CAN bus telemetry; intersections are clustered via DBSCAN ($\epsilon = 30$m), yielding 264 segments across 50 mixed-direction clusters (950 pairs).

The method ranking is directionally consistent (Table 6): V-JEPA 2 encoder-sequence DTW achieves AP = 0.867, surpassing the temporal residual (0.815) and BoT (0.791), with all DINOv3 methods at $\leq$0.623. That encoder-sequence DTW (raw encoder patches under DTW, without the residual transform) outperforms the residual confirms the comparator as primary driver, consistent with HDD; bootstrap CIs show marginal overlap ($[0.842, 0.890]$ vs. $[0.785, 0.844]$; Appendix I). The residual helps on HDD (0.956 vs. 0.942) but not nuScenes (0.815 vs. 0.867), likely because

Table 3: Context sweep on HDD (1,687 maneuvers, 64 unique frames, no padding). $\text{fps}_{\text{eff}} = 64 / d$ where d is clip duration in seconds.

context_sec	fps_{eff}	Temporal Residual AP	BoT AP	Δ (Res. − BoT)
1.0	11.1	0.962	0.924	+0.038
2.0	8.1	0.963	0.877	+0.086
3.0	6.4	0.956	0.825	+0.131
4.0	5.3	0.940	0.774	+0.166
6.5	3.8	0.872	0.663	+0.209
8.0	3.2	0.838	0.614	+0.224

nuScenes' native 2 Hz yields sparser temporal gradients where prediction errors add noise. nuScenes at native 2 Hz outperforms HDD subsampled to 2 FPS with repeat-padding (AP = 0.636), suggesting native keyframes preserve more temporal signal (Figure 8, Appendix K).

3.3 VLM Temporal Blindness (EPIC-Kitchens)

Temporal blindness in Video-LLMs is well-documented for generative QA [45, 46, 42, 52, 53]. We extend this to *retrieval embeddings*, probing where temporal signal dies. We evaluate three VLMs on 500 EPIC-Kitchens-100 [6] sequences: Qwen3-VL-8B, Gemma-4-31B, and LLaVA-Video-7B.

3.3.1 Generative Probes

Each model received identical frames with three semantically equivalent prompts asking to classify temporal direction.

Table 4: VLM forward/reverse classification on EPIC-Kitchens (500 sequences). All at chance. $\pm$ values are standard deviation across prompts.

Model	Prompt 0	Prompt 1	Prompt 2	Mean Bal. Acc.
Qwen3-VL-8B	0.517	0.502	0.504	0.508 ± 0.007
Gemma-4-31B	0.549	0.528	0.535	0.538 ± 0.009
LLaVA-Video-7B	0.503	0.501	0.505	0.503 ± 0.002

All three open-weight VLMs perform at chance. Claude 4.6 Opus [54] (200 sequences, 16 frames via API) achieves 0.545 ± 0.055, no better. Answers are driven by prompt phrasing, not temporal direction (Appendix E and F).

3.3.2 Embedding Probes: Where Temporal Signal Dies

The generative results show that VLMs *output* answers that carry no temporal information. We probe whether temporal signal is present in internal representations but discarded.

Order signal is recoverable from per-frame features but erased by pooling. Pooled VLM vision embeddings are perfectly order-invariant ($s_{\text{rev}} \approx 1.0$), but retaining per-frame tokens and comparing via temporal derivative DTW achieves perfect forward/reverse separation (balanced accuracy = 1.000). **Caveat:** this follows from the mathematical structure of temporal derivatives under DTW (reversed derivatives are negated), not learned temporal understanding. The pattern holds for both CLIP and SigLIP [39].

The LLM backbone does not surface this signal under any pooled readout ($s_{\text{rev}} = 0.998$–1.000 across all layers; Appendix B). Both logistic regression and 2-layer MLP classifiers (GroupKFold CV, fwd/rev pairs kept together) on per-position LLM hidden states with seven pooling strategies fail to classify forward vs. reverse above chance (best: 0.560; all 126 configurations within chance range; Appendix N). Yet per-position hidden states *do* retain order signal under full-sequence DTW (Appendix L). Symmetric aggregation, not the LLM backbone itself, is the bottleneck.

Table 5: Order sensitivity (s_{rev}) across the VLM pipeline on EPIC-Kitchens (500 sequences) with bootstrap 95% CIs where available. $s_{\text{rev}} \approx 1.0$ = order-invariant; lower = more order-sensitive. DINOv3 structural fingerprints shown for reference.*

Method	$s_{\text{rev}}\downarrow$ **[95% CI]**	**Bal. Acc.**$\uparrow$
DINOv3 structural fingerprints (reference)		
BoF	1.000	N/A
Chamfer	0.9999	N/A
Temporal Derivative	0.500 [0.499, 0.500]	1.000
Attention Trajectory	0.192 [0.187, 0.197]	1.000
V-JEPA 2		
BoT	0.984	0.500
Temporal Residual	0.699	0.778
VLM vision towers (pooled)		
Gemma-4 (SigLIP)	1.000†	N/A
LLaVA (CLIP)	1.000†	N/A
VLM vision sequence (per-frame DTW)		
Gemma-4 (SigLIP)	0.492 [0.491, 0.493]	1.000
LLaVA (CLIP)	0.495 [0.495, 0.496]	1.000
VLM LLM hidden state		
Gemma-4 (Gemma)	1.000 [1.000, 1.000]	N/A
LLaVA (Qwen2)	0.998 [0.998, 0.999]	N/A
Qwen3-VL (Qwen3)	0.998 [0.998, 0.999]	N/A

*DINOv3 300M params, CLIP (ViT-L) 304M, SigLIP (SO-400M) 400M. Order-invariance follows from the aggregation operator (Observation 1), not model capacity. †Raw values 1.001–1.002 from f32 precision; clamped to 1.000. Qwen3-VL vision probes are omitted from this table because its ViT uses dynamic-resolution spatial grids with 3D temporal–spatial merging, preventing clean per-frame token grouping for the detailed s_{rev} analysis; approximate pooled and sequence results appear in Table 6. The LLM hidden-state probe (below) is unaffected.

3.4 Cross-Method Summary

Table 6 consolidates results across all benchmarks; no single method dominates every paradigm.

4 Related Work

Video copy detection. Standard benchmarks (ISC [23], VSC [24], VCSL [28]) and descriptors (SSCD [25], ViSiL [26], DnS [27], S2VS [47]) provide per-frame features with learned aggregation. Wang et al. [51] win the Meta VSC challenge; subsequent analysis [7] exposes temporal-attack vulnerability. Explicit temporal alignment (VSAL [30], TransVCL [29]) shows order matters for copy localization. We diagnose *when* temporal structure helps versus hurts.

Temporal sensitivity in video models. Lei et al. [40] established the static-bias problem. Subsequent work diagnoses this in generative VLM QA [42, 41, 46, 50, 45, 53, 52]. Arrow-of-time diagnostics [16, 48] and further studies [12, 34, 17, 33, 11] show the failure is structural; PL-Stitch [9] and ArrowGEV [10] propose training-time fixes. MASH-VLM [55] identifies RoPE positional bias; TIPSv2 [56] shows scaling worsens alignment. We differ in *domain* (retrieval) and *granularity* (localizing failure to aggregation via scramble gradients and feature-vs-comparator decomposition). TGPO [35] uses binary ordered-vs-shuffled RL reward; our scramble gradient extends this to a continuous K-parameterized diagnostic. TARA [37] introduces chiral negatives for training; we use reversal as a diagnostic probe.

Video prediction for temporal understanding. V-JEPA 2 [1] learns spatiotemporal representations via masked prediction; Drive-JEPA [38] applies these to driving; TWLV-I [43] evaluates appearance vs. motion across video foundation models. Li et al. [44] show temporal information is encoded

Table 6: All methods across all tasks (VCDB: 5,585 pairs; HDD: 1,687 maneuvers; nuScenes: 950 pairs; EPIC: 500 sequences). Nymeria and MUVR scene-matching results replicate the VCDB finding (BoF dominates) and are in Appendix C. AP values are point estimates; 95% bootstrap CIs (2,000 resamples, percentile method) are in Appendix I.

Method	**VCDB (AP↑)**	**HDD (AP↑)**	**nuScenes (AP↑)**	**EPIC s_{rev}↓**	**EPIC Bal. Acc.↑**
DINOv3 BoF	0.979	0.530	0.613	1.000	N/A
DINOv3 Chamfer	**0.989**	0.559	0.612	0.9999	N/A
DINOv3 Temporal Deriv.	0.775	0.498	0.623	0.500	**1.000**
DINOv3 Attn. Trajectory	0.700	0.507	0.584	**0.192**	**1.000**
V-JEPA 2 BoT	0.838	0.825	0.791	0.984	0.500
V-JEPA 2 Encoder-Seq DTW	—	0.942	**0.867**	—	—
V-JEPA 2 Temporal Res.	0.698	**0.956**	0.815	0.699	0.778
ViCLIP (InternVid-10M)	0.907	0.542	0.618	0.996	N/A
X-CLIP (16-frame)	0.960	—	—	0.980	N/A
Optical Flow (RAFT) DTW	—	0.478	—	—	—
Qwen3-VL-8B (gen.)	0.519‡	0.531‡	0.487‡	N/A	0.508
Gemma-4-31B (gen.)	0.559‡	0.510‡	0.529‡	N/A	0.538
LLaVA-Video-7B (gen.)	0.521‡	0.500‡	0.507‡	N/A	0.503
Gemma-4 vision pooled (SigLIP)	0.926	0.521	0.662	1.000†	N/A
LLaVA vision pooled (CLIP)	0.915	0.514	0.624	1.000†	N/A
Qwen3 vision pooled (SigLIP)	0.491	0.421	0.464	0.998†	N/A
Gemma-4 vision seq (SigLIP)	0.632	0.553	0.469	0.492	**1.000**
LLaVA vision seq (CLIP)	0.637	0.545	0.470	0.495	**1.000**
Qwen3 vision seq (SigLIP)	0.610	0.468	0.517	0.542	**1.000**
VLM LLM hidden state (Gemma-4)	0.838	0.472	0.569	≥0.998	N/A
VLM LLM hidden state (LLaVA)	0.907	0.512	0.652	≥0.998	N/A
VLM LLM hidden state (Qwen3)	0.916	0.518	0.630	≥0.998	N/A

†Pooled representations are order-invariant by construction. — = not evaluated. ‡Generative balanced accuracy (forward/reverse classification); not retrieval AP. All three VLMs perform at chance. s_{rev} is N/A for generative models. VCDB AP depends on negative seed (±0.06; Appendix H).

but standard pipelines fail to surface it, matching our finding. Co-Settle [36] formalizes temporal consistency vs. semantic separability; our sensitivity–invariance analysis maps a related trade-off.

Temporal alignment for retrieval. DTW [31] remains standard; extensions include differentiable DTW [22] and Drop-DTW [49]. At $T \approx 30$ frames, each comparison takes $\sim$1 ms; top-100 reranking after FAISS [20] adds $\sim$100 ms. We use DTW as a diagnostic tool; Weller et al. [39] provide theoretical justification for multi-vector approaches.

5 Discussion

No single method dominates. **(i) Scene matching** (VCDB, Nymeria, MUVR): BoF is optimal; temporal methods add noise (Appendix C). **(ii) Motion dynamics** (Honda HDD, nuScenes): requires spatiotemporal processing (V-JEPA 2); per-frame methods and VLM vision towers all fail (AP ≤ 0.56), with the residual advantage growing under sparser sampling (Section 3.2.1); 89% of the BoT-to-residual gap comes from the comparator (cosine→DTW) on HDD, and on nuScenes the encoder-sequence DTW alone surpasses the temporal residual (AP = 0.867 vs. 0.815), confirming that the comparator is the primary driver. Even video-native ViCLIP [58] ($s_{\text{rev}} = 0.996$, HDD AP = 0.542) and X-CLIP [59] ($s_{\text{rev}} = 0.980$) do not escape the blind spot, though ViCLIP's 8-frame input partially confounds this (see Limitations). **(iii) Temporal manipulation** (reversal, scrambling): requires explicit sequence tracking; implicit 3D models [8], frame-sampling defenses [7], and Video-LLMs all remain order-invariant [9, 16]. Per-frame features separate forward/reverse under DTW (by construction), but pooling erases it; LLM hidden states stay order-invariant across all layers.

Limitations. All evaluations use pairwise similarity; DTW is not indexable ($O(T^2)$); probes train on 400 samples ($p \gg n$); ViCLIP's 8-frame input confounds its comparison with 64-frame V-JEPA 2 (Section 3.2.1 provides a partial control).

Practitioner recommendations. (1) Test your pipeline with the scramble gradient at $K \in \{1, 4, 16\}$: if AP is flat across K, the system is temporally blind. (2) For copy detection, BoF is

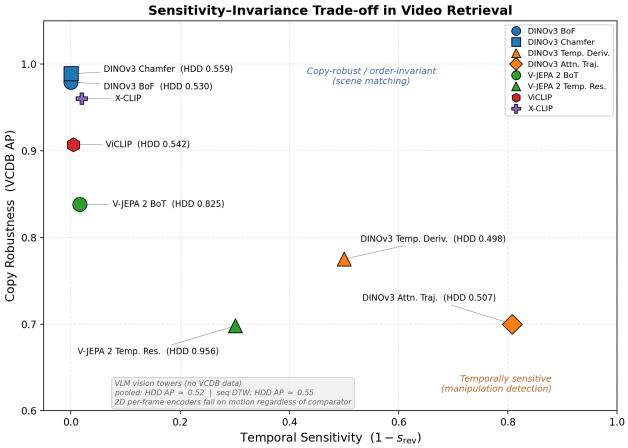

Figure 5: Sensitivity–invariance trade-off: temporal sensitivity ($1 - s_{\text{rev}}$, EPIC) vs. copy robustness (VCDB AP). No method occupies the upper-right quadrant. Note: pooled and sequence s_{rev} use different similarity functions (cosine vs. DTW) and are not directly comparable on this axis; see Eq. (1).

optimal; adding temporal methods hurts. (3) For motion-sensitive retrieval, a two-stage pipeline (BoF pre-filter $\rightarrow$ DTW rerank on top-k) is practical at $\sim$100 ms overhead per query (Appendix O).

6 Conclusion

Across six benchmarks, semantic pooling, trajectory tracking, and spatiotemporal residuals each dominate a distinct retrieval regime; no method spans both copy detection and motion retrieval (Figure 5). The temporal scramble gradient cleanly separates method classes; the feature-vs-comparator decomposition places 89% of the gap in the matching function on HDD, and on nuScenes the comparator alone (encoder-sequence DTW) surpasses the residual transform entirely (AP = 0.867 vs. 0.815); and VLM embedding probes show that pooling erases order information present in per-frame encoders (126 probe configurations at chance). Efficient order-sensitive comparators remain the key open problem.[1]

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A FPS Cap Invariance Proof

For V-JEPA 2 on Honda HDD, each clip of duration d seconds is sampled at target_fps $= 64/d$ to fill the 64-frame input. An FPS cap only activates when target_fps $>$ cap; otherwise frame selection is identical to the uncapped baseline.

Table 7 confirms that across all 9,811 HDD maneuver segments (at ± 3 s context), the maximum target_fps is 9.14. Caps at 10, 15, and 30 therefore never activate; the identical AP values at these caps are explained by frame selection invariance, not by a learned robustness plateau.

Table 7: FPS cap activation across all HDD segments (± 3 s context). Caps ≥ 10 activate for zero segments, confirming identical frame selection to the uncapped baseline. Unique frames = number of distinct video frames extracted before padding to 64.

FPS Cap	% Active	Mean Unique	Mean Dup%	Effect
2	99.9%	20.2	68.4%	Strong degradation
5	85.9%	49.1	23.3%	Moderate degradation
10	0.0%	64.0	0.0%	No-op (identical to baseline)
15	0.0%	64.0	0.0%	No-op (identical to baseline)
30	0.0%	64.0	0.0%	No-op (identical to baseline)

Segment durations range from 7.0 s to 46.0 s (mean 10.1 s), yielding target_fps $\in [1.39, 9.14]$ (mean 6.66). All values fall below the lowest non-trivial cap (10 FPS), so meaningful degradation occurs only when the cap drops below the native target rate.

B VLM Layer Ablation

To rule out the possibility that temporal order information is present at different depths of the LLM backbone, we extract hidden states from the last, middle, and penultimate layers of each VLM and compute s_{rev} (Table 8). For comparison, we include vision-token DTW ($s_{\text{rev}} \approx 0.49$), which retains per-frame temporal structure.

Table 8: s_{rev} across LLM layers for three VLM families (EPIC-Kitchens, 500 sequences). Values near 1.0 indicate order-invariance; values near 0.5 indicate perfect forward/reverse separation. Order-invariance persists across all layers in all models.

Model	Vision DTW	LLM Last	LLM Mid	LLM Penult
Qwen3-VL-8B	—*	0.998	0.998	1.000†
Gemma-4-31B	0.492	1.000	1.000†	1.000
LLaVA-Video-7B	0.495	0.998	0.998	0.999

*Qwen3-VL vision DTW omitted due to dynamic-resolution spatial grids (see Table 5). †Clamped from 1.001.

All LLM layers produce $s_{\text{rev}} \approx 1.0$: order-invariance is a property of the full LLM representation under mean-pooled readouts, not an artifact of the final layer. The temporal signal in per-frame vision tokens ($s_{\text{rev}} \approx 0.49$) is not preserved through the vision-to-LLM projection in any pooled summary.

C Full Scene Matching Results

We evaluated models across three scene-matching datasets: VCDB [3] (copy detection), Nymeria [32] (egocentric activity discrimination), and MUVR [18] (untrimmed multi-modal news/dance retrieval).

BoF and Chamfer matching consistently dominate. On Nymeria, the 100-session scale confirms that scene appearance remains the strongest signal even with high intra-class variability (AP $\approx$ 0.48). On MUVR News, matching events share visual backgrounds (e.g., news desks), favoring semantic pooling. Attention Trajectories fail because egocentric gaze and semantic pacing vary unpredictably, introducing severe intra-class noise.

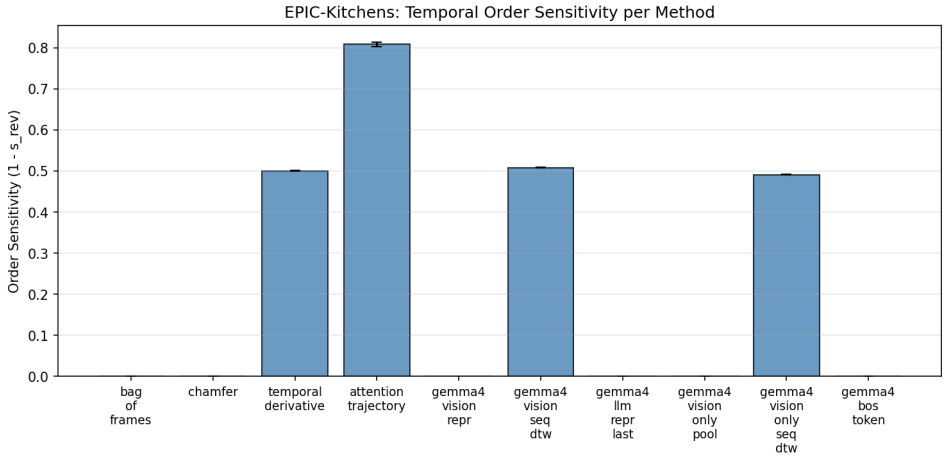

Figure 6: Order sensitivity (s_{rev}) across all embedding probes on EPIC-Kitchens (500 sequences). VLM vision towers are perfectly order-invariant when pooled ($s_{\text{rev}} \approx 1.0$), but per-frame sequences compared via DTW achieve perfect forward/reverse separation (balanced accuracy = 1.0). LLM hidden states remain order-invariant regardless.

Table 9: Retrieval AP driven by scene appearance / visual noise. VCDB: 528 videos, 5,585 positive pairs; Nymeria: 100 egocentric sessions, 8,780 segments; MUVR News: 9,958 videos, 74 topics.

Method	VCDB	Nymeria	MUVR (News)
BoF (DINOv3)	0.979	**0.485**	0.731
Chamfer matching (DINOv3)	**0.989**	0.482	**0.746**
Temporal Residual (V-JEPA 2)	0.698	0.279	0.505
Attention Trajectory (DINOv3)	0.700	0.217	0.466

D Preliminary: Qwen3-VL on VCDB

We first tested whether Video-LLMs can detect temporal reversal. Using Qwen3-VL [13], we presented forward and reversed versions of 49 VCDB videos and asked for temporal direction classification (3 trials each).

Table 10: Qwen3-VL reversal detection on VCDB. Chance = 50%.

Model	Dataset (N)	Fwd Acc.	Rev Acc.	Overall
Qwen3-VL-4B-Instruct	Single clip (1)	100%	0%	50.0%
Qwen3-VL-30B-A3B	VCDB subset (49)	83.0%	28.6%	55.8%

The 30B model achieves 55.8%, near chance. The slight improvement is driven by a content-dependent FORWARD bias (83% forward, 28.6% reverse), not temporal understanding. This motivated the systematic multi-VLM evaluation on EPIC-Kitchens (Section 3.3.1).

E VLM Temporal Integrity Probe

Separately from forward/reverse classification, we asked VLMs to judge whether a video's temporal structure is "INTACT" or "TAMPERED," on forward, reversed, and chunk-scrambled ($K = 4, 16$) versions of EPIC-Kitchens sequences.

Qwen and Gemma correctly identify forward videos as INTACT ($\sim$85–89%) and both flag reversed videos as TAMPERED (Qwen 87%, Gemma 84%). Scramble detection diverges: Qwen flags 67% of scrambled videos ($K = 4$) as TAMPERED while Gemma flags only 20%, suggesting the two models rely on different heuristics. Cross-transform consistency is very low (3.8–8.2%).

Table 11: Temporal integrity probe on EPIC-Kitchens (500 sequences): fraction judged TAMPERED. LLaVA excluded from aggregate due to constant-output degeneracy.

Model	**Forward**	**Reversed**	**Scramble** $K=4$	**Scramble** $K=16$
Qwen3-VL-8B	0.112	0.870	0.668	0.660
Gemma-4-31B	0.150	0.840	0.202	0.220
LLaVA-Video-7B†	1.000	1.000	1.000	1.000

†LLaVA produces constant "TAMPERED" regardless of input (text-only baseline confirms). Excluded from integrity analysis.

F VLM Prompt Sensitivity and Text-Only Baselines

Answers on forward/reverse classification are dominated by prompt phrasing, not visual content. Gemma-4 exhibits opposite biases across prompts: prompt 0 yields 87% reverse bias; prompt 2 yields 73% forward bias. LLaVA-Video answers "FORWARD" for prompts 0–1 (>98%) but "REVERSE" for prompt 2 (83%). Text-only baselines confirm these are prior biases; Qwen3 shows no text-only bias but still performs at chance with video.

LLaVA-Video produces constant "TAMPERED" regardless of input (text-only baseline confirms prior bias). We exclude it from aggregate integrity statistics in Appendix E.

G HDD Leakage Analysis

Since V-JEPA 2 temporal residuals achieve AP = 0.956 on Honda HDD, well above the next-best method (BoT, AP = 0.825), we verify that no metadata leakage inflates this result.

Feature inputs. All six methods receive *only* decoded video frames. GPS is used exclusively for DBSCAN clustering; timestamps select the frame window (± 3 s) and are discarded before extraction.

Segment duration. Maneuver segments span 7.0–46.0 s (mean 10.1 s). Left and right turns have similar duration distributions (left: 10.3 ± 3.8 s; right: 9.9 ± 3.6 s), ruling out duration as a shortcut. V-JEPA 2 pads all segments to 64 frames, eliminating length signal.

Cluster composition. Only "mixed" clusters containing both turn directions are used. Within each, all $\binom{n}{2}$ pairs are evaluated, ensuring models must distinguish maneuver type at the same location.

Cross-session control. To rule out session adjacency confounds, we re-evaluate excluding same-session pairs (91.4% of pairs are already cross-session). AP is unchanged: V-JEPA 2 temporal residual 0.957 (cross-session) vs. 0.956 (all pairs); BoF 0.536 vs. 0.530. This confirms results are not inflated by within-session visual correlations.

Mechanistic explanation. The context sweep (Section 3.2.1) shows temporal residual AP degrades monotonically as context narrows, consistent with genuine temporal processing. BoF methods remain near chance across all windows.

Motion proxy check. To rule out trivial motion magnitude as a shortcut, we compute mean pixel difference between consecutive frames for each segment and correlate pairwise proxy similarity with V-JEPA 2 residual similarity across all 97,731 pairs. Spearman $r = 0.051$; the motion proxy itself achieves only AP = 0.474 (below chance). Residuals capture motion *structure*, not intensity.

H VCDB Negative-Sampling Sensitivity

Our primary VCDB evaluation uses 1:1 positive-to-negative sampling (5,585 pairs each). The full dataset contains $\binom{528}{2} \approx 139$K possible pairs. We re-evaluate at three ratios to verify robustness (Table 12).

Table 12: VCDB AP across negative-sampling ratios. *Rank order and relative gaps are stable across all settings.* All-pairs uses the full $\binom{528}{2}$ set (requiring $\sim$140K DTW comparisons).

Method	**1:1**	**1:5**	**All pairs**
DINOv3 Chamfer	0.989	0.949	0.845
DINOv3 BoF	0.979	0.929	0.829
DINOv3 Temporal Deriv.	0.713	0.424	0.228
DINOv3 Attn. Trajectory	0.700	0.408	0.225

Rank order (Chamfer > BoF > Temporal Deriv. > Attn. Trajectory) is preserved across all ratios. AP decreases with more negatives as expected; AUC remains stable (e.g. Chamfer AUC: 0.990, 0.989, 0.989). The 1:1 temporal derivative AP (0.713) differs from Table 6 (0.775) because this analysis uses independently sampled negatives with a different seed; rank order is unaffected.

I Bootstrap Confidence Intervals

Table 13 reports 95% bootstrap confidence intervals (2,000 resamples, percentile method, seed = 42) for the headline AP values from Tables 1–3. Degenerate bootstrap samples (all-positive or all-negative) fall back to the point estimate.

J HDD Qualitative Analysis

To complement the quantitative maneuver discrimination results, we present qualitative evidence that V-JEPA 2 temporal residuals capture genuine motion dynamics rather than exploiting a visual shortcut.

Nearest-neighbor examples. For three representative intersection clusters, the top-3 neighbors by temporal residual similarity retrieve the same maneuver type despite identical backgrounds; errors occur when both turns share similar speeds through the same intersection phase.

Background similarity statistics. Within each GPS cluster, all segments share the same intersection. BoF cosine similarity for all $\binom{n}{2}$ pairs per cluster: mean 0.63 ± 0.18 (same-direction: 0.63 ± 0.18; cross-direction: 0.60 ± 0.16; gap = 0.029), confirming appearance is non-discriminative. V-JEPA 2 temporal residual similarity aligns with maneuver labels (same: 0.008 ± 0.006; cross: 0.003 ± 0.001). Figure 7 shows the score distributions.

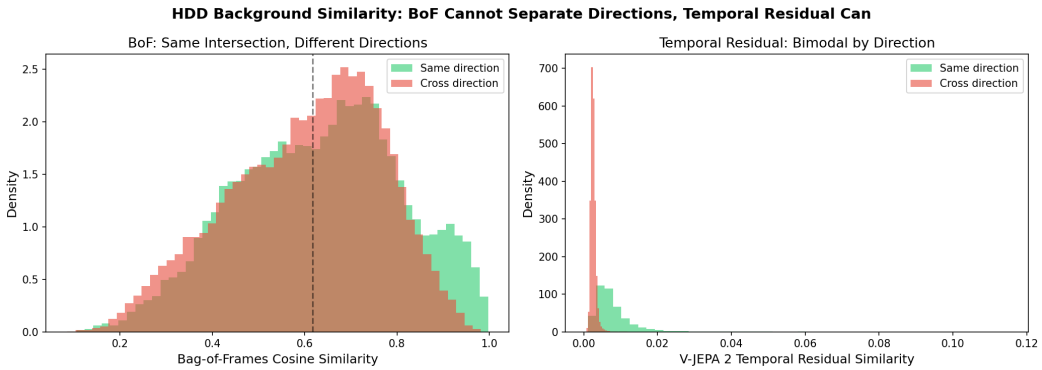

Figure 7: Pairwise similarity distributions on HDD intersection clusters. BoF similarities overlap heavily regardless of maneuver match, while V-JEPA 2 temporal residual scores separate same- vs. cross-direction pairs.

Table 13: 95% bootstrap CIs for AP across all benchmarks. "Width" is $\mathrm{CI}_{\mathrm{high}} - \mathrm{CI}_{\mathrm{low}}$. VCDB and HDD CIs are narrow (<0.02); nuScenes, Nymeria, and MUVR CIs are wider due to smaller evaluation sets.

Benchmark	Method	AP	95% CI	Width
VCDB	DINOv3 Chamfer	0.989	[0.987, 0.991]	0.004
	DINOv3 BoF	0.979	[0.976, 0.981]	0.005
	DINOv3 Temporal Deriv.	0.775	[0.765, 0.784]	0.019
	DINOv3 Attn. Trajectory	0.700	[0.689, 0.711]	0.022
	V-JEPA 2 BoT	0.838	[0.829, 0.846]	0.017
	V-JEPA 2 Temporal Res.	0.698	[0.687, 0.709]	0.022
HDD	V-JEPA 2 Temporal Res.	0.956	[0.955, 0.958]	0.003
	V-JEPA 2 BoT	0.825	[0.821, 0.828]	0.007
	DINOv3 Chamfer	0.559	[0.555, 0.564]	0.009
	DINOv3 BoF	0.529	[0.525, 0.534]	0.009
	DINOv3 Attn. Trajectory	0.507	[0.502, 0.512]	0.010
	DINOv3 Temporal Deriv.	0.498	[0.494, 0.503]	0.009
nuScenes	V-JEPA 2 Encoder-Seq DTW	0.867	[0.842, 0.890]	0.048
	V-JEPA 2 Temporal Res.	0.815	[0.785, 0.844]	0.059
	V-JEPA 2 BoT	0.791	[0.761, 0.820]	0.059
	DINOv3 Temporal Deriv.	0.623	[0.580, 0.665]	0.085
	DINOv3 BoF	0.613	[0.570, 0.658]	0.088
	DINOv3 Chamfer	0.612	[0.570, 0.659]	0.089
	DINOv3 Attn. Trajectory	0.584	[0.540, 0.634]	0.094
Nymeria	DINOv3 BoF	0.485	[0.482, 0.489]	0.007
	DINOv3 Chamfer	0.482	[0.478, 0.485]	0.007
	DINOv3 Temporal Deriv.	0.274	[0.271, 0.277]	0.006
	DINOv3 Attn. Trajectory	0.217	[0.215, 0.219]	0.004
	V-JEPA 2 BoT	0.420	[0.417, 0.424]	0.007
	V-JEPA 2 Temporal Res.	0.278	[0.276, 0.281]	0.005
MUVR News	DINOv3 BoF	0.731	[0.724, 0.740]	0.016
	DINOv3 Chamfer	0.745	[0.738, 0.754]	0.016
	DINOv3 Temporal Deriv.	0.476	[0.468, 0.485]	0.017
	DINOv3 Attn. Trajectory	0.466	[0.458, 0.475]	0.017
	V-JEPA 2 BoT	0.597	[0.587, 0.605]	0.018
	V-JEPA 2 Temporal Res.	0.505	[0.496, 0.514]	0.018

VCDB and HDD CIs are narrow (<0.025) under pair-level resampling. nuScenes CIs are wider (0.06–0.09) due to the smaller evaluation set (950 pairs), but rank order remains unambiguous. Nymeria CIs are very tight (<0.01) despite the scene-retrieval setting (394,210 pairs). MUVR News CIs (0.016–0.018) reflect the moderate evaluation set (27,453 pairs). **Note:** pair-level resampling underestimates CI width because pairs within a GPS cluster share clips. A cluster-level block bootstrap (resampling clusters with replacement) produces CIs 7–32$\times$ wider on HDD: e.g., V-JEPA 2 temporal residual widens from $[0.955, 0.958]$ to $[0.890, 0.978]$. Rank order is preserved; the pair-level CIs should be read as lower bounds on uncertainty.

K nuScenes Cross-Dataset Visualization

L LLM Hidden-State Probe: Vision-Token-Only Analysis

A potential objection to our LLM hidden-state probe is that mean-pooling over *all* tokens reintroduces symmetric aggregation. To address this, we extract LLM hidden states restricted to *vision-token positions only*. Three variants:

1. **Vision-only mean pool**: Mean-pool LLM hidden states at vision-token positions only. Compare via cosine similarity.

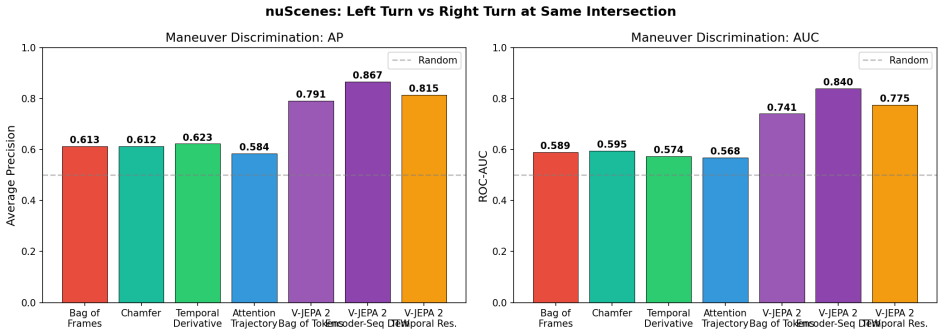

Figure 8: nuScenes cross-dataset validation (264 segments, 950 pairs, 50 intersection clusters). V-JEPA 2 encoder-sequence DTW (AP = 0.867) is the strongest method; the comparator change alone surpasses the temporal residual (AP = 0.815).

2. **Vision-only per-frame sequence**: Group vision-token hidden states by their source frame, pool per frame to obtain a (T, D) sequence. Compare via temporal derivative DTW.
3. **BOS/first token**: Use the first token's hidden state (BOS or system token) as a summary representation. Compare via cosine similarity.

Table 14: s_{rev} for vision-token-only LLM hidden-state probes (EPIC-Kitchens, 500 sequences). If vision-only pooling recovers temporal signal, we expect $s_{\text{rev}} \ll 1.0$; if order-invariance persists, the conclusion from Section 3.3.2 is strengthened.

Model	Vision-only pool	Vision-only seq DTW	BOS token
Qwen3-VL-8B	0.998 [0.998, 0.998]	**0.542** [0.540, 0.544]	0.996
Gemma-4-31B	1.000 [1.000, 1.000]†	**0.509** [0.509, 0.509]	1.000
LLaVA-Video-7B	0.999 [0.998, 0.999]	**0.517** [0.516, 0.518]	1.000†

Vision-only mean pooling remains $s_{\text{rev}} \approx 1.0$ across all models, confirming that symmetric aggregation, not the LLM backbone, erases temporal signal. The per-frame sequence probe yields low s_{rev} (balanced accuracy = 1.0), showing that LLM hidden states preserve per-position order information that pooling discards. BOS tokens are order-invariant, indicating the model's summary position captures no temporal signal. †Raw values >1.0 from f32 precision; clamped to 1.000.

M Scramble Gradient Data

Table 15: AP across scramble levels K on VCDB (527 of 528 videos; one decode failure). $K = 1$ is the unscrambled baseline. Order-invariant methods show constant AP; sequence-aware methods show overall degradation (multi-seed analysis: $\sigma \leq 0.007$; Figure 9).

Method	$K=1$	$K=2$	$K=3$	$K=4$	$K=6$	$K=8$	$K=12$	$K=16$
BoF (DINOv3)	0.979	0.979	0.979	0.979	0.979	0.979	0.979	0.979
Chamfer (DINOv3)	0.989	0.989	0.989	0.989	0.989	0.989	0.989	0.989
V-JEPA 2 BoT	0.838	0.838	0.838	0.838	0.838	0.838	0.838	0.838
Temporal Derivative (DINOv3)	0.775	0.735	0.733	0.734	0.724	0.731	0.729	0.722
V-JEPA 2 Temporal Residual	0.698	0.612	0.633	0.609	0.601	0.618	0.609	0.639
Attention Trajectory (DINOv3)	0.700	0.650	0.631	0.619	0.598	0.596	0.585	0.572

V-JEPA 2 residuals show non-monotonic degradation (e.g. $K = 3$ higher than $K = 2$), likely due to boundary effects where the spatiotemporal prediction window crosses chunk boundaries. A multi-seed analysis (10 independent shuffle seeds per K) shows that order-invariant methods have zero variance ($\sigma = 0.000$), while temporal methods show small but real variation ($\sigma \leq 0.007$) that does not affect method ranking (Figure 9). Because the main scramble gradient shuffles *extracted* embeddings, it tests whether the comparator (DTW) is sensitive to temporal order, not whether

V-JEPA's encoder responds differently to scrambled input. A raw-frame scramble (re-running V-JEPA 2 on chunk-shuffled video) produces clean monotonic degradation: residual AP drops from 0.705 at $K = 1$ to 0.652 at $K = 16$, while BoT remains flat at 0.839. The encoder does produce different residuals from temporally disrupted input.

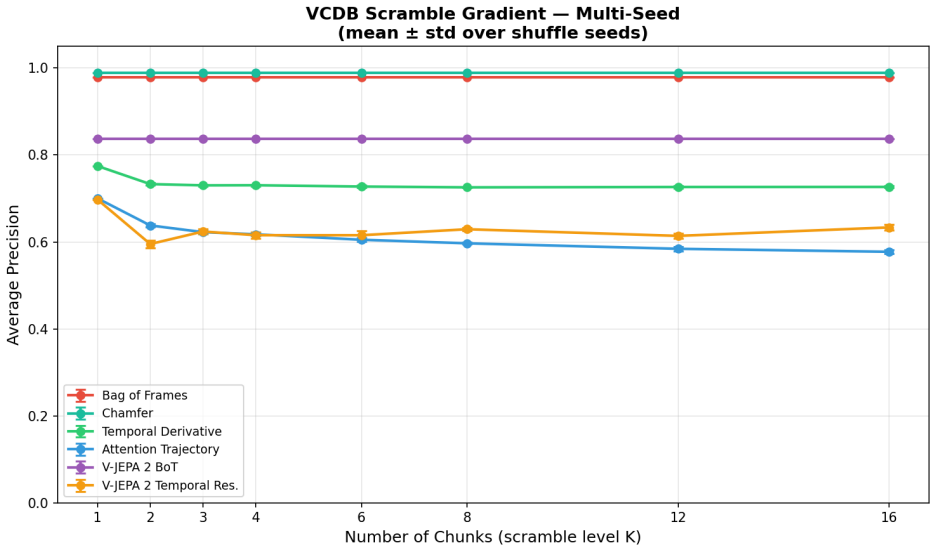

Figure 9: Multi-seed scramble gradient on VCDB (10 seeds per K, mean $\pm$ std shaded). Order-invariant methods (BoF, Chamfer, V-JEPA 2 BoT) have zero variance. Temporal methods show small variance ($\sigma \leq 0.007$) that does not change the ranking.

N Linear Probe on LLM Hidden States

The s_{rev} analysis (Section 3.3.2) shows that mean-pooled LLM hidden states are order-invariant. A natural follow-up: is temporal signal present in the per-position hidden states but lost to symmetric aggregation? If so, a learned classifier should recover it.

We extract last-layer hidden states from all three VLMs for 200 EPIC-Kitchens sequences (forward and reverse), yielding 400 samples per model. We train ℓ_2-regularized logistic regression (5-fold GroupKFold CV, fwd/rev pairs kept together, $C \in \{0.01, 0.1, 1, 10, 100\}$) on seven readout strategies:

- **mean**: mean over all positions (replicates s_{rev} finding)
- **max**: element-wise max over all positions
- **first / last**: single position (BOS or final token)
- **concat_first_last**: concatenation of first and last positions
- **vision_mean**: mean restricted to vision-token positions
- **vision_concat_fl**: first and last vision-token positions

Neither linear nor nonlinear probes recover temporal order from LLM hidden states. All 126 configurations (3 models $\times$ 7 pooling strategies $\times$ {5 regularization strengths for linear, 1 for MLP}) land between 0.458 and 0.560, within the range expected from chance on 400 samples. The "first" token gives exactly 0.500 under both probes, a sanity check that the pipeline is correct when the input is truly identical across conditions (BOS tokens do not change between forward and reverse).

Note that per-position LLM hidden states *do* carry order signal when compared as a sequence via DTW ($s_{\text{rev}} \approx 0.51$–$0.54$; Table 14): order is encoded in the relative content across positions, but no fixed-dimensional reduction of the sequence retains it. This is the same pooling-destroys-order dynamic observed at the vision encoder level, replicated one layer deeper.

Table 16: Probe accuracy (GroupKFold CV, fwd/rev pairs kept together) for forward vs. reverse classification from LLM hidden states. Linear = logistic regression (best across $C \in \{0.01, 0.1, 1, 10, 100\}$); MLP = 2-layer, 256 hidden units, ReLU. Chance = 0.500. No configuration exceeds chance.

	Gemma-4 31B		**LLaVA-Video 7B**		**Qwen3-VL 8B**	
Pooling	**Linear**	**MLP**	**Linear**	**MLP**	**Linear**	**MLP**
mean	0.512	0.552	0.500	0.505	0.537	0.510
max	0.532	0.458	0.470	0.490	0.528	0.502
first	0.500	0.500	0.500	0.500	0.500	0.500
last	0.510	0.507	0.515	0.495	0.550	0.512
concat_first_last	0.515	0.490	0.505	0.478	0.555	0.520
vision_mean	0.520	0.550	0.502	0.512	0.537	0.505
vision_concat_fl	0.528	0.500	0.495	0.467	0.515	0.535

O Computational Costs

Table 17: Per-video feature extraction and per-pair comparison costs on a single H200 GPU. T = number of frames per clip.

Method	**Extraction (ms/video)**	**Comparison (ms/pair)**
DINOv3 BoF ($T = 30$)	$\sim$120	$<$0.01 (cosine)
DINOv3 Chamfer ($T = 30$)	$\sim$120	$\sim$0.3 (frame matrix)
DINOv3 Attn. Trajectory ($T = 30$)	$\sim$150	$\sim$1 (DTW)
DINOv3 Temporal Deriv. ($T = 30$)	$\sim$120	$\sim$1 (DTW)
V-JEPA 2 BoT ($T = 64$)	$\sim$280	$<$0.01 (cosine)
V-JEPA 2 Temporal Res. ($T = 64$)	$\sim$560	$\sim$1 (DTW)
VLM vision pooled	$\sim$800–2,500	$<$0.01 (cosine)
VLM LLM hidden state	$\sim$1,500–5,000	$<$0.01 (cosine)

V-JEPA 2 temporal residual requires both encoder and predictor forward passes, roughly doubling extraction cost vs. BoT. DTW comparison at $T \approx 30$ is $\sim$1 ms; a top-100 reranking pass after FAISS [20] pre-filtering adds $\sim$100 ms total, making two-stage retrieval practical for production workloads.

P Dataset and Model Licenses

All experiments use publicly available datasets and pretrained models. Table 18 lists the governing license for each asset. Use is restricted to non-commercial research, consistent with the most restrictive license among the assets combined in any given experiment. No new datasets or model weights are released by this paper.

Honda HDD access required a signed Data Use Agreement with HRI-USA; nuScenes required account registration at nuscenes.org; other datasets are downloadable under their respective non-commercial licenses. All assets were used for non-commercial research evaluation only.

Table 18: Licenses for datasets and pretrained model weights used in this work. Verbatim license text is available from each asset's canonical distribution channel; see `REPRODUCIBILITY.md` for HuggingFace IDs and download URLs.

Asset	**License**	**Distribution**
Datasets		
VCDB [3]	Research-only; no warranty (Fudan Univ.)	Project page
Honda HDD [4]	HRI-USA Data Use Agreement (non-commercial)	HRI-USA
nuScenes [5]	CC BY-NC-SA 4.0	nuscenes.org
EPIC-Kitchens-100 [6]	CC BY-NC 4.0	epic-kitchens.github.io
Nymeria [32]	CC BY-NC 4.0	facebookresearch GH
MUVR [18]	Research-only (authors' terms)	Authors' release
Pretrained model weights		
DINOv3 [2]	Meta FAIR non-commercial	HuggingFace
V-JEPA 2 [1]	CC BY-NC 4.0	HuggingFace
ViCLIP / InternVid [58]	Apache 2.0 (code); CC BY-NC 4.0 (weights)	OpenGVLab
X-CLIP [59]	MIT	microsoft/VideoX
Qwen3-VL-8B [13]	Apache 2.0	HuggingFace
Gemma 4 [14]	Gemma Terms of Use	HuggingFace
LLaVA-Video [15]	Apache 2.0 (code); Llama-2 terms (weights)	HuggingFace
RAFT (optical flow)	BSD-3-Clause	princeton-vl/RAFT